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Roll No.

Degree: B-Tech (ECAM)

Semester: III — Course work

MID-SEMESTER EXAMINATION, September, 2025

Course Title: DIGITAL CIRCUITS & SYSTEMS

Course Code: EAEPC305

Duration: 1Hr 30 Minutes

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Find the Octal, BCD, Excess-3 and Hexadecimal equivalents of $(23)_{10}$	2	I
1b	If $x = (14.75)_{10}$ and $y = (9.25)_{10}$, convert x and y to binary and find their difference $(x - y)$ using 2's complement method.	2	I
2a	If $p = 724$ and $q = 539$, find the BCD sum of p and q.	2	I
2b	Using truth table, prove that complement of $(\bar{A}.B + A.\bar{B})$ is equal to $(\bar{A}.\bar{B} + A.B)$	2	I
3a	Design a 3 bit Binary to Gray code converter.	2	I, II
3b	Design a Half Subtractor, implement using only universal gates.	2	II
4a	Minimize using k-map: $F(A,B,C,D) = \sum_{\min} (2, 8, 12, 14) + d(4, 5, 6, 10)$	2	II
4b	Draw and explain the logic diagram of a single digit BCD adder.	2	II
5a	Design an 8:1 multiplexer, using two 4:1 multiplexers.	2	II
5b	Implement using 8:1 Multiplexer: $F(w, x, y, z) = \sum_{\min} (0, 1, 6, 7, 9, 12, 13, 14)$	2	II